Speechology

This is a deep learning Text-to-Speech (TTS) model designed to be used with Augmentative and Alternative Communication (AAC) software

Members

DeAnna Griffith

Problem

The problem I want to address is the quality and naturalness of TTS systems used in common AAC software to help support language development with children who need AAC's.

Why It Matters

Many AAC systems rely on synthesized speech to allow users to communicate an essentially unlimited range of words and sentences. However, synthesized voices can sometimes produce incorrect or unnatural pronunciations, particularly for less common words, names, or context-dependent pronunciations. The voices may also sound noticeably different from natural human speech. For some children they may never be able to fully communicate without the help of a device, but many have their speech development supported by the use of an AAC. This is why it's important to make sure that those pronunciations are correct, because it is one of the main facets in which they learn to speak themselves.

While some AAC systems address pronunciation and naturalness by using pre-recorded speech, pre-recorded messages limit the number of things a child can communicate and make it difficult to generate new, complex sentences. So, it works better for younger children who will communicate with more simplicity, but becomes less viable as the child gets older.

This creates a tradeoff between flexibility and natural-sounding speech. Synthesized speech provides the flexibility to say virtually anything, while pre-recorded speech can sound more natural but is limited to previously recorded words or phrases.

Proposed Solution

My proposed solution is to use deep learning to develop a more natural-sounding TTS system specifically for AAC applications. The goal would be to preserve the flexibility of synthesized speech—allowing a child to generate virtually any sentence—while producing speech that is more natural and human-like.

Data Sets

There are many different datasets specifically for TTS in deep learning. Most (if not all) I could find utilized a supervised model, matching the spoken audio to the synched text transcript. One dataset I came across was LJSpeech. It has around 13,100 data pairs (sound files and transcripts), and uses one speaker for all the files, with around 24 hours of audio. This would be beneficial for creating a consistent voice. However, it also seems to use a lot of higher level vocabulary, which for my project wouldn't be necessary, since I would want to focus on communication for middle to elementary aged kids.

Multilingual LibriSpeech was another dataset I found that also offered soundbytes in other languages, acting as an extension of the LibriSpeech dataset which is in English which has about 1000 hours of data. I could utilize them in conjecture to potentially add a multilingual compatibility. This is something that I was thinking of since the overall scope of my original project seems like it wouldn't be too difficult considering the large amounts of research that already exist on the topic. I could also start with just the English files, and if I have the capability to add other languages, do that after the fact. The biggest stipulation is that LibriSpeech was originally created with the intention of turning audio into text. The dataset could be used to do the reverse, but may require some extra steps.

The last dataset I looked at was Libri-TTS/ LibriTTS-R. It uses multiple different speakers, varying in accent. The audio has a total runtime of about 585 hours. It also has the capability to view the audio in context, meaning I could use it to mimic that cadence in the final product. This also means that I would have more diverse data to work with, but that it would be more challenging to get the best end result.

Given all of this information, I'm thinking either LibriSpeech or Libri-TTS/ LibriTTS-R would be the best option. While LibriSpeech has more hours of data, as well as the multilingual addon, the audio quality isn't as high as Libri-TTS/ LibriTTS-R. LibriSpeech, as I mentioned, also wasn't built for turning text into audio, which is the focus of my project. Libri-TTS/ LibriTTS-R also already has the preprocessing done for you. So, it is for these reasons that the dataset I will focus on will be **Libri-TTS/ LibriTTS-R**.